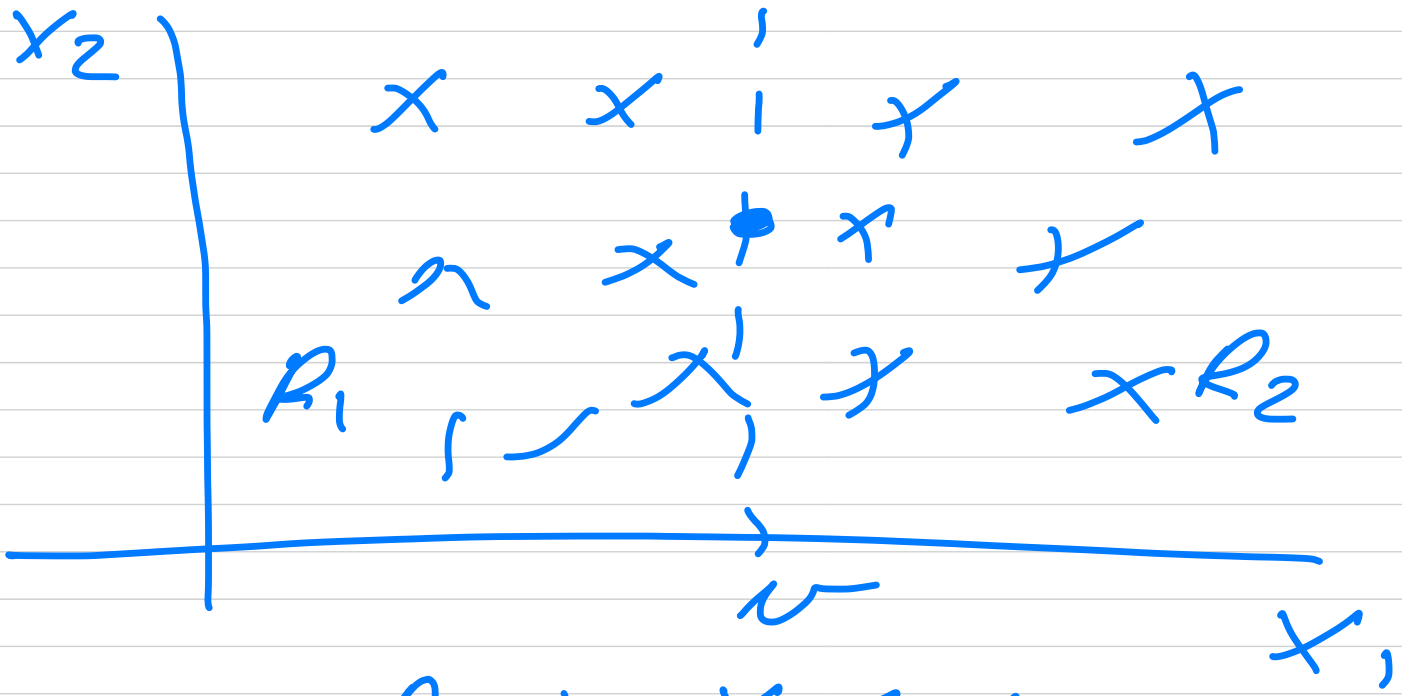
FYS-STK3155/4155 November 9

FYS-STK3155/4155 November 9

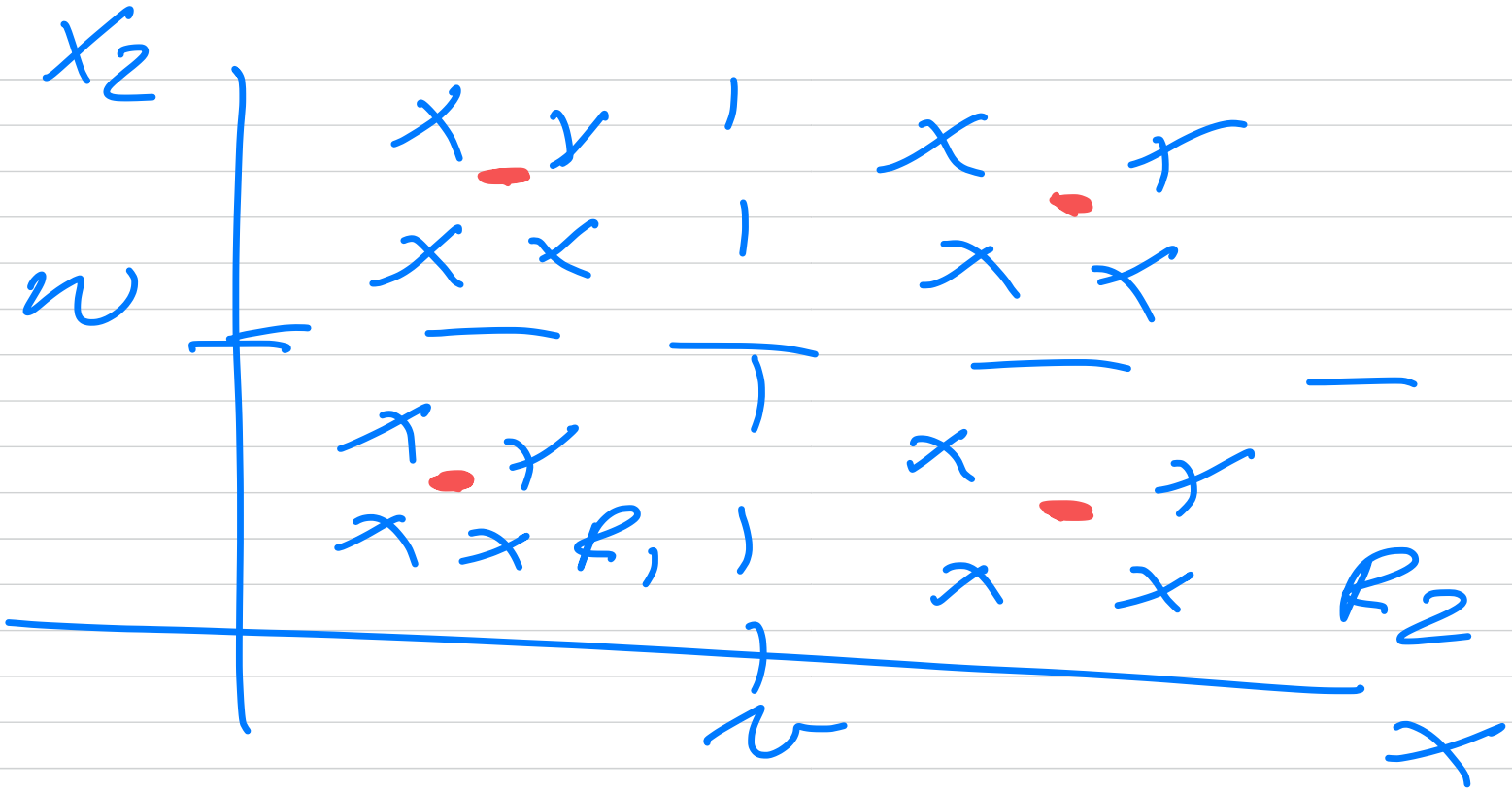
Regression problem

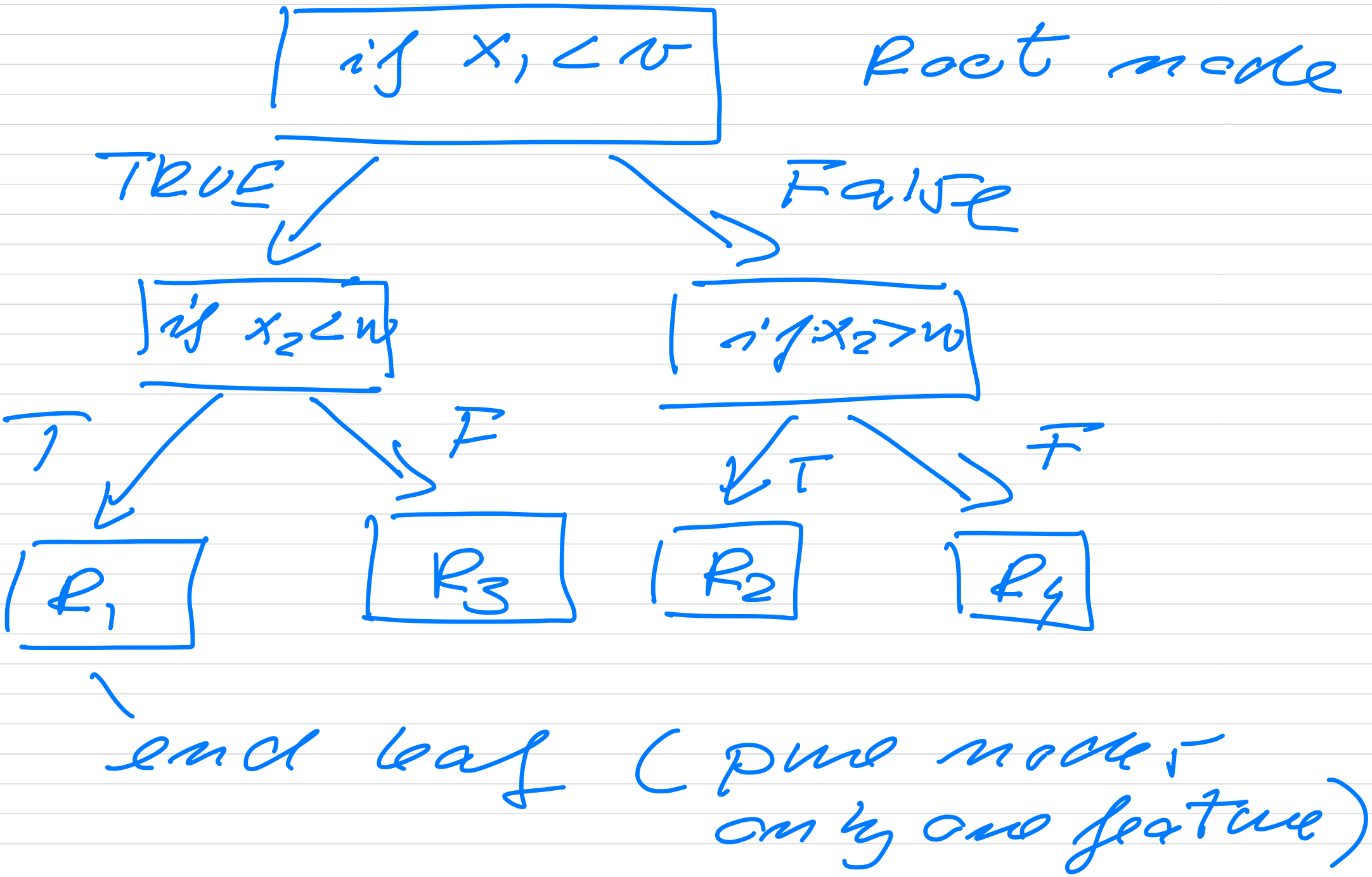


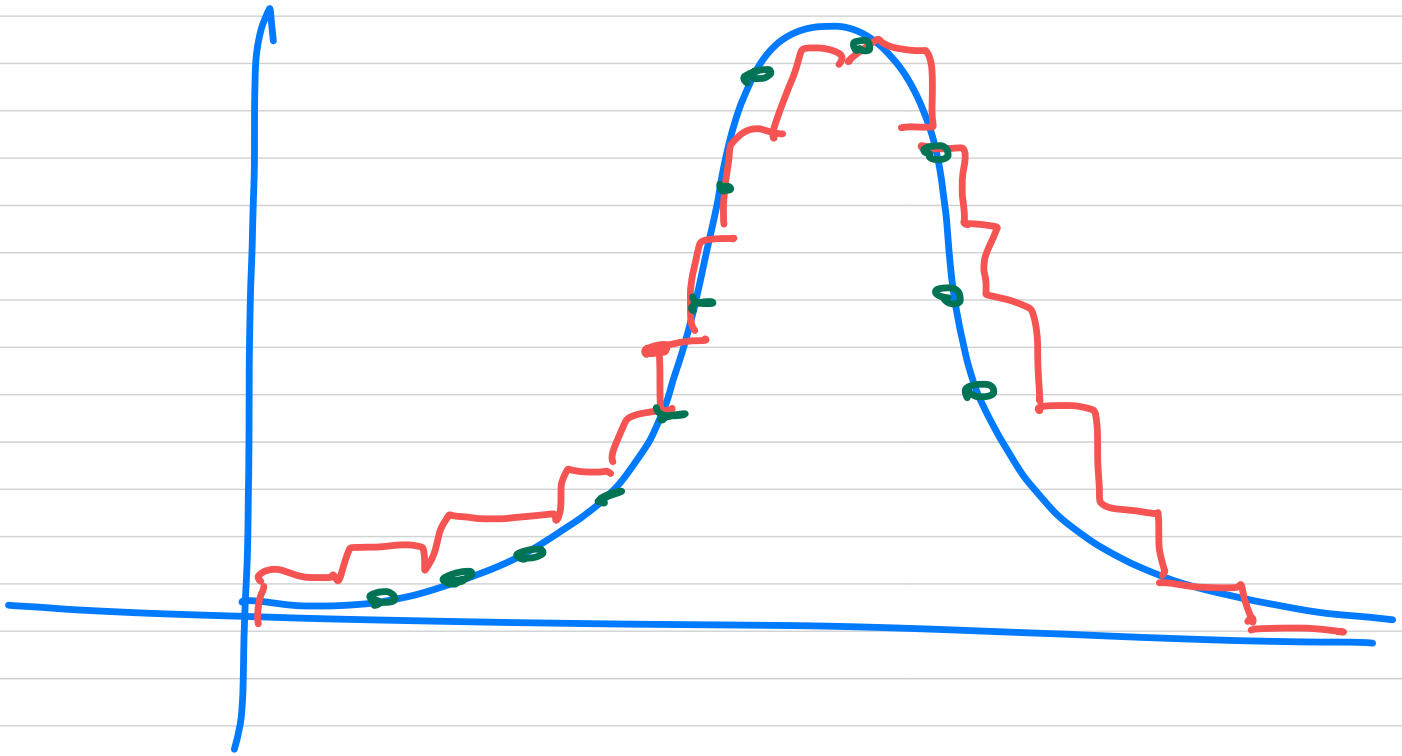
R_1 : $x_1 \leq v$

R_2 ; $x_1 > v$

prediction on x_1 is the mean value of R_1 and R_2


x_2
R_1
R_2


if $x_1 < v$
Root node
TRUE
False
if $x_2 < w$
if $x_2 > w$
T
F
T
F
R_1
R_3
R_2
R_4
end leaf (pure node; only one feature)



How do we select the branches?

- Gini
- Entropy

$g, s \in [0, 1]$

$g = 0$ pure case

$s = 0$ — " —

Example : Binary case

probability $p \in [0, 1]$

True $= p$

False $= 1 - p$

$$\sum_{i=1}^{2} p_i = (1-p) + p = 1$$

Criterion for g and s^-

- continuous
- non-negative
- high value for unlikely events
- additive

entropy

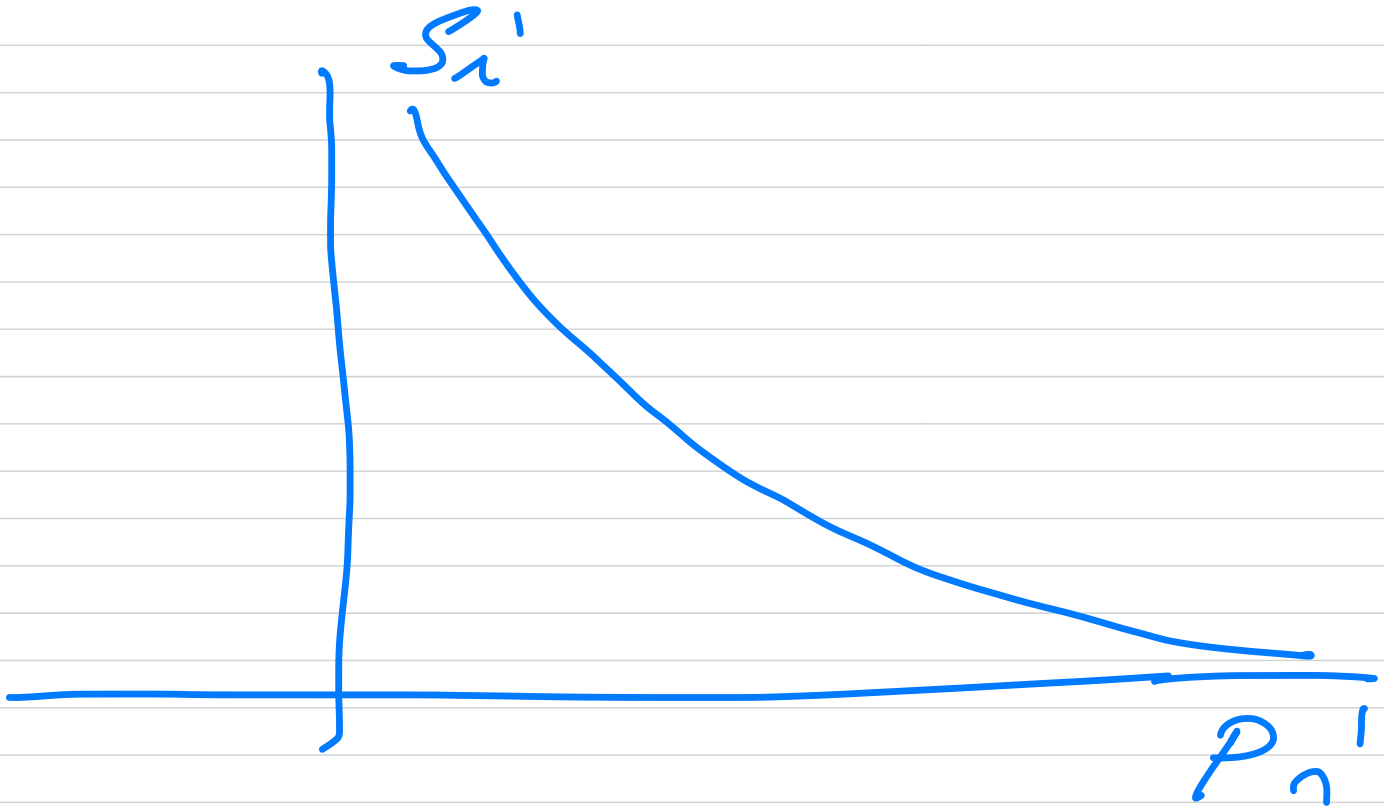
$$s_i' = -p_i' \log_2 p_i'$$

why not $s_i' = -\log_2 p_i'$

$$P_i' = p(x_1)\,p(x_2)$$

$$-\log P_i' = -\log p(x_1) - \log p(x_2)$$

$$= S_1'$$



$$S = -\sum_{i=1}^{2} p_i \log_2 p_i$$

p, $1-p$ ($0 \log_2 0 = 0$)

$$S = -p \log_2 p - (1-p) \log_2 (1-p)$$

True — False

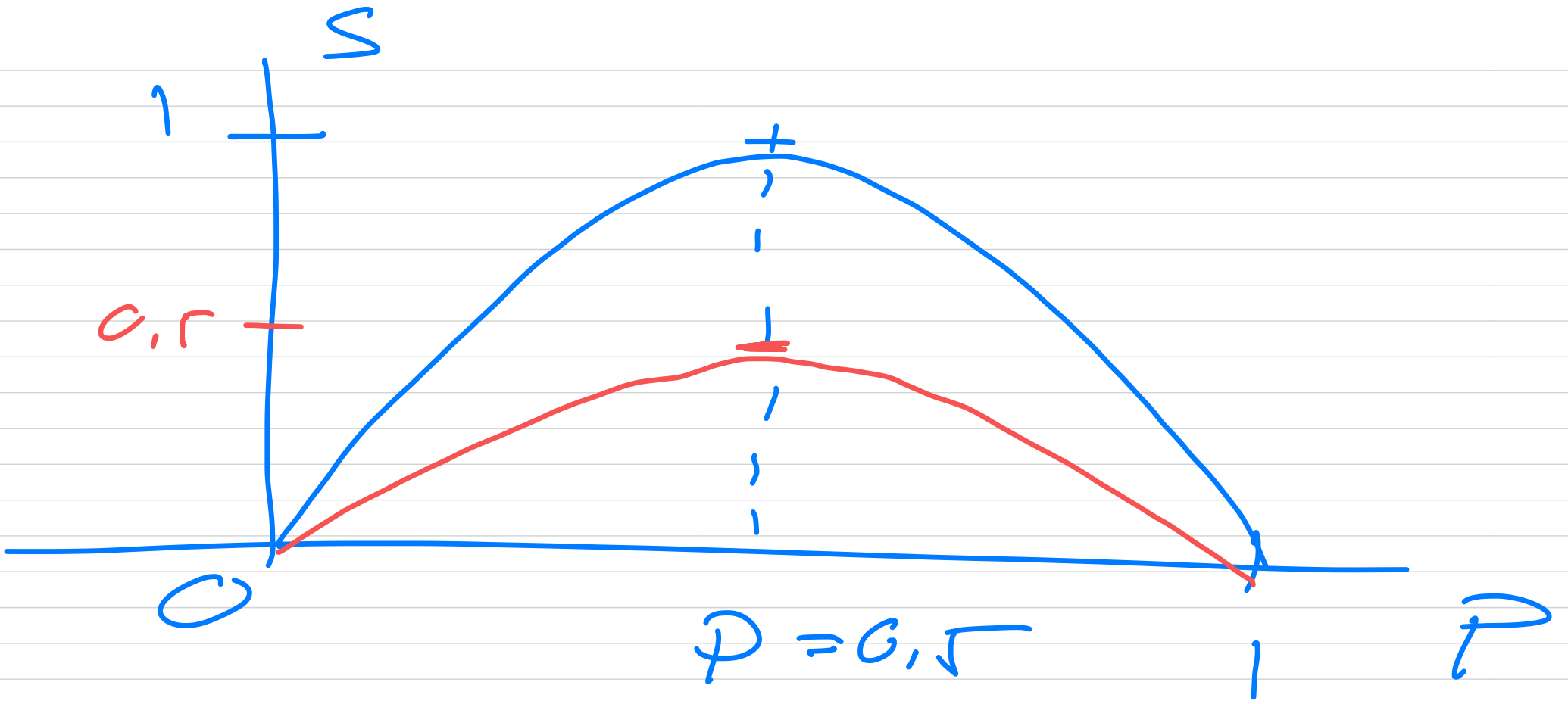
$S = 0$ if $p = 0$ False

$S = 0$ if $p = 1$ True

True, pure

pure case

only one possibility



Gini' factor

$$g = \sum_{i=1}^{M} p_i' (1 - p_i')$$

$$= \sum_i p_i' - \sum_i p_i^2 = 1 - p_i^2$$

$$\geq 1$$

$p = 0$
$g = 0$

$p = 1$ $g = 0$

$p = 0,5$
$g = 0,5$

Example

inputs
- Average grade
  A(bove), B(elow)
- hours studied
  H(igh), L(ow)
- hours slept
  H, L

output
- Grade in course
  A(bove), B(elow)

Trend	study	sleep	Grade
A	L	H	A
B	H	L	B
A	L	H	A
A	H	H	A
B	L	H	B
A	L	L	B
B	H	H	B
B	L	H	B
A	L	L	B
A	H	H	A

$P(Trend = A) = 6/10$

$P(Trend = B) = 4/10$

if Trend = A ∩ grade B

$P = 2/6 = 1/3$

if Trend = A ∩ grade A

$P = 4/6$

$gini = 1 - \left(\left(\frac{4}{6}\right)^2 + \left(\frac{2}{6}\right)^2\right) = 0.45$

if Trend = B ∩ grade = Above

$P = 0$

if Trend = B ∩ grade = Below

$Gini = (1 - (1^2 + 0)) = 0$ $P = 4/4 = \underline{1}$

weighted sum for trees

$6/10 \times 0.45 + 4/10 \cdot 0 = \boxed{0.27}$

gini for hours slept

$g = \boxed{0.47}$

Gini for hours studied

$g = \boxed{0.34}$

Choose smallest one for root.

Root

Grade Trend
g = 0.27
[6,4]

Trend above

Trend below

?
LHS

?

LHS	Sleep	Grade	Study
A	L	H	A
A	L	H	A
A	H	H	A
A	L	L	B
A	L	L	B
A	H	H	A

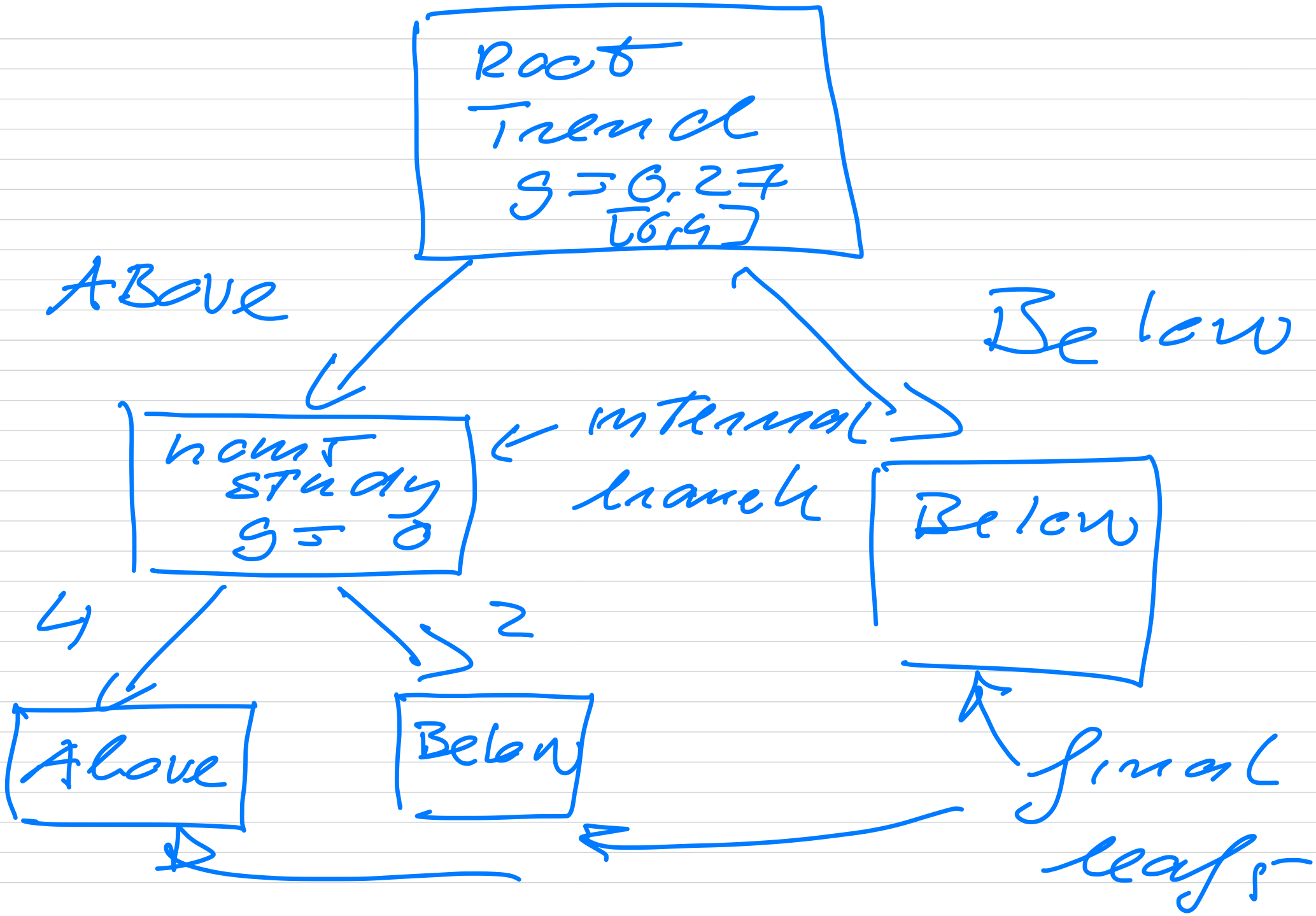
→

Simi for sleep
0.33

Simi for study
1

RHS –

Trench	sleep	study	grade
B	H	L	B
B	L	H	B
B	H	H	B
B	L	H	B


Root
Trend
S=0.27
[0,9]
Above
Below
mount
study
S= 0
internal
branch
Below
4
2
Above
Below
final
leafs